AMP Call Tracker

Backend Technical Architecture, Database Models & API Routes

Based on Official Business Requirements Document (BRD) | Database: MySQL 8.0 Stored Procedures

1. System & Architecture Overview

The AMP Call Tracker backend is engineered using Node.js and Express in TypeScript. It is designed around 100% Stored Procedures on MySQL 8.0 with zero raw/inline SQL queries executed from application code, delivering strict SQL injection immunity, pre-compiled query plan performance, and role-based data isolation. The system supports multi-project tracking, contact deduplication by mobile number, rapid call outcome logging, follow-up schedule tracking (Today/Overdue/Upcoming), user target monitoring, and Excel (.xlsx) report generation.

2. Database Models (Relational Tables)

2.1 Model: users (User Accounts, Roles & Targets)

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INC	Unique User Identifier
full_name	VARCHAR(255)	NOT NULL	Full display name of user / caller
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email address
mobile_number	VARCHAR(20)	UNIQUE, NOT NULL	Login & contact phone number
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
role	ENUM('admin','user')	NOT NULL, DEFAULT 'user'	Role authorization level
status	ENUM('active','inactive')	NOT NULL, DEFAULT 'active'	Account activity state
daily_target	INT	NULL, DEFAULT 0	Target calls per day set by Admin
weekly_target	INT	NULL, DEFAULT 0	Target calls per week set by Admin
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

2.2 Model: projects (AMP Initiatives & Campaigns)

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INC	Unique Project Identifier
name	VARCHAR(255)	NOT NULL	Name of AMP campaign / project
description	TEXT	NULL	Detailed overview and calling brief
start_date	DATE	NULL	Campaign commencement date
end_date	DATE	NULL	Campaign target end date
status	ENUM('active','closed')	NOT NULL, DEFAULT 'active'	Project lifecycle state
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

2.3 Model: project_users (Agent Assignment to Projects)

Field Name	Data Type	Constraints	Description
project_id	INT	FK -> projects(id)	Associated campaign/project
user_id	INT	FK -> users(id)	Assigned calling agent / user
assigned_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of assignment

2.4 Model: contacts (Master Contact Database)

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INC	Unique Contact Identifier
name	VARCHAR(255)	NOT NULL	Contact full name
mobile_number	VARCHAR(20)	UNIQUE, NOT NULL	Primary mobile (used for deduplication)
alternate_mobile	VARCHAR(20)	NULL	Secondary contact telephone
email	VARCHAR(255)	NULL	Contact email address
city	VARCHAR(100)	NOT NULL	Geographic city
state	VARCHAR(100)	NOT NULL	Geographic state
pincode	VARCHAR(10)	NOT NULL	Postal ZIP code
category	ENUM	NOT NULL	'Donor', 'Beneficiary', 'Student', 'Volunteer', 'Other'

Field Name	Data Type	Constraints	Description
assigned_user_id	INT	FK -> users(id), NULL	Calling agent currently assigned
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record ingestion timestamp

2.5 Model: project_contacts (Multi-Project Contact Mapping)

Field Name	Data Type	Constraints	Description
project_id	INT	FK -> projects(id)	Associated campaign/project
contact_id	INT	FK -> contacts(id)	Associated contact record
added_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Mapping inclusion timestamp

2.6 Model: calls (Call Log Records)

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INC	Unique Call Record Identifier
user_id	INT	FK -> users(id)	Caller who performed the call
contact_id	INT	FK -> contacts(id), NULL	Linked contact record if known
project_id	INT	FK -> projects(id), NULL	Project under which call was placed
contact_number	VARCHAR(20)	NOT NULL	Dialed phone number
duration_seconds	INT	NOT NULL, DEFAULT 0	Actual call duration captured via SIM log
status	ENUM('answered','not_answered')	NOT NULL	Call pickup status (duration > 0)
call_status	ENUM	NULL	'Connected', 'Not Answered', 'Busy', 'Switched Off', 'Call Back Later', 'Wrong Nu
interest_level	ENUM	NULL	'Interested', 'Maybe', 'Not Interested' (only if Connected)
purpose	VARCHAR(255)	NULL	Note / Short discussion summary
outcome	ENUM	NULL	'interested', 'not_interested', 'unanswered', 'follow_up_required'
attempt_number	INT	NOT NULL, DEFAULT 1	Sequential attempt count for contact/project
follow_up_date	DATETIME	NULL	Date & time scheduled for next follow-up
called_at	DATETIME	NOT NULL	Phone clock timestamp call was placed
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	System sync timestamp

2.7 Model: follow_ups (Follow-up Tasks & Reminders)

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INC	Unique Follow-up Identifier
call_id	INT	FK -> calls(id)	Originating call log record
contact_id	INT	FK -> contacts(id)	Target contact to call back
user_id	INT	FK -> users(id)	Agent responsible for follow-up
due_at	DATETIME	NOT NULL	Scheduled reminder date and time
status	ENUM	NOT NULL, DEFAULT 'pending'	'pending', 'completed', 'rescheduled'
rescheduled_to	DATETIME	NULL	Updated datetime if rescheduled
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Record creation timestamp

3. Database Stored Procedures Specification

- sp_user_create: Register / Admin create user with email, mobile, password hash, role
- sp_user_find_by_email / sp_user_find_by_email_or_mobile: Authentication credential checks
- sp_user_get_all: Admin listing of all callers with active status and assigned targets
- sp_user_update_status / sp_user_update_targets: Manage user activation and daily/weekly quotas
- sp_project_create / sp_project_update / sp_project_list: Manage multi-project initiatives
- sp_project_assign_user / sp_project_get_users: Project membership bindings
- sp_contact_create_batch / sp_contact_create_single: Deduplicated contact intake (ON DUPLICATE KEY UPDATE)
- sp_contact_get_assigned: Agent view of personal contacts with keyword search and category filtering
- sp_contact_get_context: Auto-populates pop-up header (attempt count, last note, last status, pending follow-up)
- sp_contact_assign_or_reassign: Reassigns contact to another agent while retaining call history
- sp_call_create: Ingests completed call, auto-increments attempt count, records duration
- sp_call_update: Updates note, outcome, interest level, and schedules follow-up date
- sp_call_get_paginated: Filtered call history (date range, project, user, status, outcome)
- sp_followup_get_user_queue: Categorizes user follow-ups into Today (yellow), Overdue (red), Upcoming
- sp_followup_mark_done / sp_followup_reschedule: Updates follow-up task state
- sp_dashboard_user_stats: Total calls, connected, interest breakdown, targets vs achievement

- sp_dashboard_admin_stats: Project-wise, user-wise performance, overdue follow-ups team totals
- sp_reports_export_raw: Pre-joined tabular dataset for Excel (.xlsx) streaming

4. REST API Routes Specification

4.1 Authentication & Profile Routes (/api/auth)

POST	/api/auth/login	*[Public]*	Authenticates user via email/mobile + password, returns JWT with role
GET	/api/auth/me	*[Authenticated]*	Retrieves current logged-in user profile, role, and active targets
POST	/api/auth/change-password	*[Authenticated]*	Updates user password verifying current password hash

4.2 Admin User Management Routes (/api/admin/users)

GET	/api/admin/users	*[Admin Only]*	Lists all users with search, role, status, and daily/weekly quotas
POST	/api/admin/users	*[Admin Only]*	Creates a new caller or admin account
PATCH	/api/admin/users/:id/target	*[Admin Only]*	Sets or updates daily and weekly calling quotas for a user
PATCH	/api/admin/users/:id/status	*[Admin Only]*	Activates or deactivates a user account

4.3 Admin Project Management Routes (/api/admin/projects)

GET	/api/admin/projects	*[Admin Only]*	Lists all AMP projects/campaigns with status and assigned agent count
POST	/api/admin/projects	*[Admin Only]*	Creates a new campaign with start/end dates and brief
PATCH	/api/admin/projects/:id	*[Admin Only]*	Updates project metadata, dates, or marks project as Closed
POST	/api/admin/projects/:id/users	*[Admin Only]*	Assigns multiple calling agents to a project
GET	/api/admin/projects/:id/users	*[Admin Only]*	Lists all calling agents assigned to a specific project

4.4 Contacts & Assignment Routes (/api/contacts, /api/admin/contacts)

GET	/api/contacts/my	*[Agent/User]*	Fetches assigned contacts for caller with fast name/mobile search and category filter
GET	/api/contacts/:id/context	*[Agent/User]*	Auto-fills Log Call pop-up: attempt count, last note, last status, pending follow-up
GET	/api/contacts/:id/history	*[Agent/User]*	Retrieves complete historical chronological call log and notes for a contact
GET	/api/admin/contacts	*[Admin Only]*	Lists all master contacts filtered by project, assignment, city, or category
POST	/api/admin/contacts/import	*[Admin Only]*	Batch uploads contacts from Excel/CSV with automatic mobile number deduplication
POST	/api/admin/contacts/assign	*[Admin Only]*	Assigns or reassigns single/multiple contacts to a user and project

4.5 Calls & Outcome Logging Routes (/api/calls)

POST	/api/calls	*[Agent/User]*	Logs raw completed call from SIM dialer (contact_number, duration, called_at)
PATCH	/api/calls/:id	*[Agent/User]*	Updates call pop-up: note, call_status, interest_level, follow_up_date
GET	/api/calls	*[Authenticated]*	Paginated call log list with search, status, outcome, project, and date filters
GET	/api/calls/:id	*[Authenticated]*	Returns full single call detail including linked contact and follow-up

4.6 Follow-up Tasks Routes (/api/follow-ups)

GET	/api/follow-ups	*[Agent/User]*	Fetches caller's follow-up queue categorized into Today, Overdue, and Upcoming
PATCH	/api/follow-ups/:id/done	*[Agent/User]*	Marks a scheduled follow-up reminder as completed
PATCH	/api/follow-ups/:id/reschedule	*[Agent/User]*	Reschedules follow-up to a newly chosen date and time

4.7 Dashboards & Analytics Routes (/api/dashboards)

GET	/api/dashboards/user	*[Agent/User]*	Personal stats: Total calls, Connected, Interested, Target vs Achievement, Due Today
GET	/api/dashboards/admin	*[Admin Only]*	Team stats with Project, User, and Date-range filters; highlights team overdue items

4.8 Excel Export & Reports Routes (/api/admin/reports)

GET	/api/admin/reports/date-wise	*[Admin Only]*	Streams Excel (.xlsx) of day-by-day calling volumes and outcomes
GET	/api/admin/reports/project-wise	*[Admin Only]*	Streams Excel (.xlsx) of project comparison, contact reach, and interest rates
GET	/api/admin/reports/user-wise	*[Admin Only]*	Streams Excel (.xlsx) of caller productivity, talk time, and target compliance
GET	/api/admin/reports/interested	*[Admin Only]*	Exports verified list of Interested / Maybe contacts with notes for campaign leads
GET	/api/admin/reports/follow-ups	*[Admin Only]*	Exports pending and overdue follow-ups with responsible agent names